World, Requirements, and Specifications (WRS)

Theia: Accessible Indoor Navigation for Blind and Low-Vision Users

CptS 484 Team 10 {Placeholder}

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Revision History

Version	Date	Author	Description
0.1	October 11, 2026	Danbi Kim	Drafted initial outline, introduction, and section 2.
0.1.1	October 12, 2026	Alex Shaw	Reviewed the team's progress, regenerated the baseline, and delegated tasks for on-time submission.
0.2	October 12, 2026	Danbi Kim	Wrote section 2.1 and 2.2.
0.3	October 15, 2026	Danbi Kim	Wrote section 2.3 and 2.4.
0.4	October 17, 2026	Seth Trimberger	Wrote section 4.1 and 4.2.
0.5	October 17, 2026	Alex Shaw	Added onto sections 2.2 and 2.3. Wrote sections 3, 4.4, 5, and Appendix A.
0.6	October 17, 2026	Johanna Siebol	Edited grammar and made suggestions on section 4.2 and 4.3.
0.7	October 18, 2026	Seth Trimberger	Added app mockups and mockups descriptions.
0.8	October 18, 2026	David Del Val Fraire	Revised section 4.3 as user-facing instructions. Aligned emergency handoff and countdown behavior with Figma. Recorded presentation.
0.9	October 18, 2026	Alex Shaw	Added references and table of contents. Proofread and refined deliverables prior to submission.
0.9.1	October 18, 2026	Johanna Siebol	Edited some grammar issues in section 2.1, 2.2, 4.1 and 4.2.

0.9.2	October 18, 2026	Danbi Kim	Fixed grammar issues and refined all across the section.

1. Introduction

Theia is a proposed native iOS application intended to help blind and low-vision users navigate between rooms and other significant locations inside a building. The application is expected to communicate location and route information through accessible interaction, including spoken guidance and platform accessibility features.

The project scope is limited to indoor, room-to-room or location-to-location navigation. Navigation within individual rooms and detection or avoidance of moving obstacles such as pedestrians are outside the defined scope.

The purpose of this WRS is to clarify the world and stakeholder context, resolve important uncertainties in the preliminary definition, specify a manageable set of functional and non-functional requirements, and establish traceability from real-world problems to the software requirements.

2. Issues with Preliminary Definition Given

2.1 Issues with II.1 Domain, Stakeholders, Functional and Non-Functional Objectives

2.1.1 Meaning of significant location

Issue description: The project plan refers to rooms and other significant locations, but the exact set of destinations is not defined. It is not clear whether entrances, exits, stairs, elevators, restrooms, and so on are included.

Options:

- Option A: Limit the prototype to named rooms.
- Option B: Include an enumerated set of rooms and shared destinations.

Provisional decision and rationale: Support named rooms and a small set of named building destinations only when represented in the prototype's location data.

2.1.2 Target-user variation:

Issue description: "Blind and low-vision users" describes a diverse group. The definition may not clarify whether users rely on VoiceOver, enlarged text, audio, haptics, or combinations of them.

Options:

- Option A: Optimize only for screen-reader users.
- Option B: Provide a screen-reader first interaction while preserving basic visual accessibility for low-vision users.

Provisional decision and rationale: Support VoiceOver-compatible interaction and readable,

high-contrast visual controls where shown in the prototype. Do not assume all users have the same abilities.

2.1.3 Meaning of emergency dial-out

Issue description: Emergency dial-out is described as required, but the trigger and whether the app calls a local emergency number directly or merely opens the phone dialer are not specified.

Options:

- Option A: Place a call automatically.
- Option B: Open the system dialer with the emergency number and require the user to initiate the call.

Provisional decision and rationale: For Phase I, demonstrate a handoff to the native Phone app when the user requests help or an unanswered fall alert reaches 30 seconds. The user completes the calling steps in the Phone app. Testing uses mock interactions without placing live emergency calls.

2.1.4 Definition of accessibility success

Issue description: The plan references WCAG 2.2, but a mobile native application also relies on platform accessibility behavior. It is unclear which criteria or test cases the team will use.

Options:

- Option A: Claim general compliance.
- Option B: Define a small, testable set of accessibility checks for the prototype.

Provisional decision and rationale: Use specific checks for VoiceOver labels and order, text legibility, control accessibility, and understandable audio instructions.

2.2 Issues with II.2 Software System Requirements: Functional Requirements

2.2.1 Current-location update and beacon interpretation

Issue description: The project plan says the prototype simulates beacon signals but does not define the event format, what happens when a signal is missing, or whether repeated signals cause repeated announcements.

Options:

- Option A: Treat every signal as a location update.
- Option B: Define a simulated event format and behavior for duplicate, unknown, and missing signals.

Provisional decision and rationale: The prototype will use predefined location identifiers. Recognized identifiers update the current location, unknown identifiers produce a recoverable status message, and duplicate events will not announce the same location unless the user requests it.

2.2.2 Destination selection and route availability

Issue description: The expected way to choose a destination and the system response when no route exists are not clear in the plan.

Options:

- Option A: Only demonstrate one fixed route.
- Option B: Allow the user to select from a small list and show a clear result when no route is defined.

Provisional decision and rationale: Support selecting a destination from a small predefined list and demonstrate at least one valid route. If no route exists, the application must announce that it cannot provide a route.

2.2.3 Route instruction content

Issue description: The level of detail and format for route guidance are not defined. A direction could be ambiguous if it lacks information.

Options:

- Option A: Give general directions.
- Option B: Provide concise step-by-step instructions using available named locations and decision points.

Provisional decision and rationale: Provide concise step-by-step instructions based on the predefined route data and announce arrival when the simulated location matches the destination.

2.2.4 Emergency dial-out workflow

Issue description: The required emergency capability lacks a defined user flow, confirmation step, and testing boundary.

Options:

- Option A: Automatic calling.
- Option B: Open the native Phone app on a help request or after an unanswered fall-alert countdown, leaving calling controls in Phone.

Provisional decision and rationale: The prototype will demonstrate the native Phone app handoff after a help request or an unanswered 30-second fall-alert countdown. The native calling interface handles subsequent calling steps. Tests use mock interactions.

2.2.5 Vertical navigation and floor transitions

Issue description: The plan does not specify how Theia represents floors, gives floor-transition instructions, or confirms that the user reached the floor.

Options:

- Option A: Limit the prototype to one floor.

- Option B: Include a small multi-floor map with a simulated floor transition and user-operated elevator/stair use.

Provisional decision and rationale: Limit the Phase I prototype to single-floor navigation. Multi-floor navigation remains a planned capability for Phase II, requiring additional specifications for floor transitions, elevator and stair guidance, and floor confirmation.

2.2.6 Static obstacle identification

Issue description: The prototype includes guidance around a pillar, but it is unclear whether the system detects obstacles using sensors or retrieves their positions from predefined building data.

Options:

- Option A: Represent known static obstacles in predefined route data.
- Option B: Detect obstacles dynamically using device sensors.

Provisional decision and rationale: Use predefined static obstacle locations for the prototype. This supports route guidance without requiring computer vision or additional hardware. Moving obstacle detection remains outside the project scope.

2.2.7 Fall detection and emergency escalation

Issue description: Fall assistance needs separate rules for requesting help, dismissing a false alarm, and reaching the end of an unanswered countdown. Opening the native Phone app must be distinguished from placing or connecting a call.

Options:

- Option A: Simulate a fall alert that opens the native Phone app when the user requests help or when a 30-second countdown expires without a response.
- Option B: Implement automatic emergency calling after an unanswered countdown.

Provisional decision and rationale: Use Option A for Phase I, matching the Figma workflow. “Call for help” opens the Phone app immediately. “I’m okay” dismisses the alert and cancels the countdown. If the user does not respond within 30 seconds, the same Phone app handoff occurs. The user completes calling steps in the Phone app. Demonstrations use mock interactions without placing live emergency calls.

2.3 Issues with II.3 software system requirements: Non-functional Requirements

2.3.1 Measurable accessibility and usability

Issue description: Accessibility is a stated success driver but the plan does not define measurable acceptance checks or methods.

Options:

- Option A: State that the app will be accessible.
- Option B: Define checks that reviewers can observe in the prototype.

Provisional decision and rationale: Evaluate VoiceOver discoverability, meaningful labels, and

instructions that do not rely on color or vision alone.

2.3.2 Response time and feedback

Issue description: The plan does not specify how quickly a location event or user action should produce feedback.

Options:

- Option A: Leave timing unspecified.
- Option B: Define a prototype-level target.

Provisional decision and rationale: For the prototype, provide feedback within 5 seconds after a user action or simulated event under normal conditions. This is a proposed target, not a claim about the performance.

2.3.3 Privacy and data handling

Issue description: The plan anticipates on-device operation without a cloud backend, but does not define sharing of location or data collection.

Options:

- Option A: Assume data may be stored or transmitted.
- Option B: Minimize collection and avoid network transmission in the prototype.

Provisional decision and rationale: The prototype will use local, simulated data and will not transmit location or data to a server.

2.3.4 Platform and device constraints

Issue description: The plan specifies a native iOS app, but minimum iOS version, supported devices, audio conditions, and required permissions are not stated.

Options:

- Option A: Support multiple mobile platforms.
- Option B: Constrain the prototype to an iPhone/iOS design.

Provisional decision and rationale: Target an iPhone/iOS interaction model and will be evaluated in the prototype environment selected by the team.

2.3.5 Navigation safety during location loss

Issue description: The preliminary definition does not specify whether navigation continues when the application cannot determine the user's current location. Continuing to issue directions using outdated location information could be unsafe.

Options:

- Option A: Continue providing directions using the last known location.
- Option B: Pause navigation and inform the user that location information is unavailable.

Provisional decision and rationale: Use Option B. The application will instruct the user to stop

walking, pause route guidance, and offer an option to locate the user again or stop navigation.

3. World, Requirements, and Specifications (WRS)

3.1 W

3.1.1 Problems

P-01: Blind and low-vision users may have difficulty identifying the correct distance and location for turns while navigating unfamiliar buildings.

P-02: Room locations and directions may not be available in formats accessible to users who rely on voice interaction or screen readers.

P-03: Known static obstacles, such as walls and pillars, may create navigation hazards when their locations are not communicated.

P-04: Indoor positioning may become unavailable or inaccurate, preventing users from knowing whether they remain on the intended route.

P-05: Users experiencing falls or other emergencies may have difficulty requesting assistance promptly.

P-06: Navigation software may expose sensitive information if user location or activity data is unnecessarily transmitted or retained.

3.1.2 Goals

G-01: Allow users to select indoor destinations through accessible visual or voice interactions.

G-02: Provide understandable, step-by-step spoken directions that account for known static obstacles.

G-03: Allow users to operate and review navigation without relying entirely on visual interaction.

G-04: Communicate location uncertainty and provide safe recovery options when navigation cannot continue.

G-05: Provide accessible emergency assistance controls and demonstrate the intended emergency workflow.

G-06: Minimize collection, storage, and transmission of user information.

3.1.3 Improved understanding of II.1

Theia targets blind and low-vision iPhone users navigating between named rooms and other significant indoor locations.

The primary stakeholders are users requiring accessible navigation. Secondary stakeholders include building operators responsible for location information, emergency service providers involved in the calling workflow, and the development team responsible for implementation and testing.

The system operates using a predefined building model containing room identifiers, route connections, decision points, and known static obstacles. The prototype simulates wireless

beacon location updates rather than requiring physical beacon installation.

Users may select destinations through a room list or supported spoken commands. The application communicates navigation instructions through speech and accessible interface elements.

The Phase I prototype focuses on single-floor, room-to-room navigation. Navigation within individual rooms, moving obstacle detection, and real-world emergency service integration are outside Phase I validation.

Multi-floor navigation remains a planned capability but is deferred beyond Phase I. Floor-transition guidance and confirmation will require additional specification before implementation.

3.2 RS

3.2.1 Functional RS: Improved understanding of II.2 functional requirements

FR-01 [Must]: The system shall display a selectable list of predefined destinations.

FR-02 [Must]: The system shall accept a spoken destination identifier and request confirmation of the recognized destination.

FR-03 [Must]: The system shall provide an option to retry or manually select a destination when speech recognition fails.

FR-04 [Must]: The system shall update its current location when receiving a recognized simulated beacon identifier. Repeated identical identifiers shall not cause duplicate location announcements unless requested.

FR-05 [Must]: The system shall identify unknown or unavailable location signals and provide a recoverable status message.

FR-06 [Must]: The system shall determine whether a route exists between the current location and selected destination. If no route exists, the system shall notify the user.

FR-07 [Must]: The system shall obtain a valid simulated current location before beginning route guidance.

FR-08 [Must]: The system shall provide spoken, step-by-step directions using predefined route information and identifiable decision points.

FR-09 [Must]: The system shall provide guidance around known static obstacles represented in the route data.

FR-10 [Must]: The system shall announce arrival when the simulated current location matches the selected destination.

FR-11 [Must]: The system shall allow users to stop or cancel navigation.

FR-12 [Must]: The system shall provide accessible voice-command feedback and allow users to repeat the current instruction.

FR-13 [Must]: The system shall provide an accessible emergency assistance action that opens the native Phone app on the user’s request. Phase I testing shall demonstrate the handoff

through mock interactions without placing live emergency calls.

FR-14 [Must]: The prototype shall demonstrate a simulated fall alert with a 30-second countdown. “Call for help” shall initiate the native Phone app handoff immediately. “I’m okay” shall dismiss the alert and cancel the countdown. If the user does not respond before the countdown expires, the prototype shall demonstrate the same Phone app handoff. The user remains responsible for completing the calling steps in Phone. No live emergency calls shall be placed during Phase I testing.

FR-15 [Must]: The system shall pause navigation, instruct the user to stop walking, and provide recovery options when the current location becomes unavailable.

A requirement for multi-floor navigation remains pending until the team defines floor-transition behavior and acceptance criteria.

3.2.2 Non-functional RS: Improved understanding of II.3 non-functional requirements

NFR-01 [Must]: All primary interactive controls shall have meaningful accessibility labels and a logical VoiceOver navigation order.

NFR-02 [Must]: Important instructions and statuses shall be communicated without relying exclusively on color or visual information. The interface shall use readable, high-contrast text.

NFR-03 [Must]: The prototype shall provide feedback within five seconds of supported user actions and simulated location events under normal test conditions.

NFR-04 [Must]: Navigation and simulated location processing shall operate on-device without transmitting location information to a remote server.

NFR-05 [Must]: Emergency calling demonstrations shall use mock interactions rather than actual calls to emergency services.

NFR-06 [Must]: Navigation failures shall produce understandable feedback and provide a recovery or cancellation option.

NFR-07 [Must]: The prototype shall use an iPhone/iOS interaction model supporting the selected Apple accessibility features.

NFR-08 [Must]: The prototype shall distinguish simulated positioning and safety capabilities from verified real-world functionality.

Verification will use manual interface walkthroughs, simulated location events, VoiceOver inspection, and observation of expected system responses. These checks establish prototype-level acceptance rather than full WCAG 2.2 certification.

4. Preliminary Prototype and User Manual

Theia's preliminary prototype consists of 11 interface mockups that were created in Figma that demonstrate the flow of the mobile app and how users would use the app based on 4 TO-BE scenarios. The design was informed by accessibility guidance for blind and low vision users including large text and controls, strong color contrast, generous line spacing, and consistent section placement. Microphone and speaker icons indicate when the app is listening or

speaking.

4.1 Prototype Overview

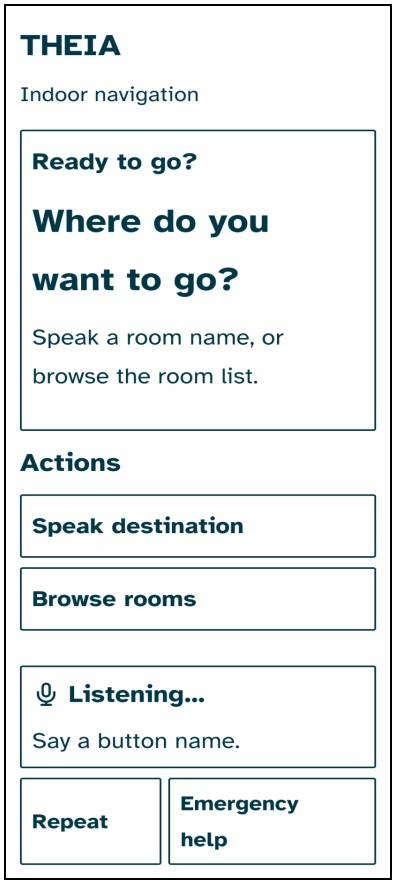
Theia's preliminary prototype consists of 11 iPhone interface mockups demonstrating accessible indoor navigation and emergency assistance.

The interface prioritizes spoken instructions, recognizable actions, high-contrast text, and simple navigation states. Users can select a destination, receive route guidance, request repeated instructions, and respond to simulated navigation failures.

The prototype assumes predefined building locations and simulated beacon signals. It illustrates the expected user experience rather than demonstrating verified real-world positioning, obstacle sensing, or emergency calling.

The mockups cover 3 principal workflows: destination selection, active navigation, and emergency assistance. The complete interface mockups are available in the project's Figma design [1].

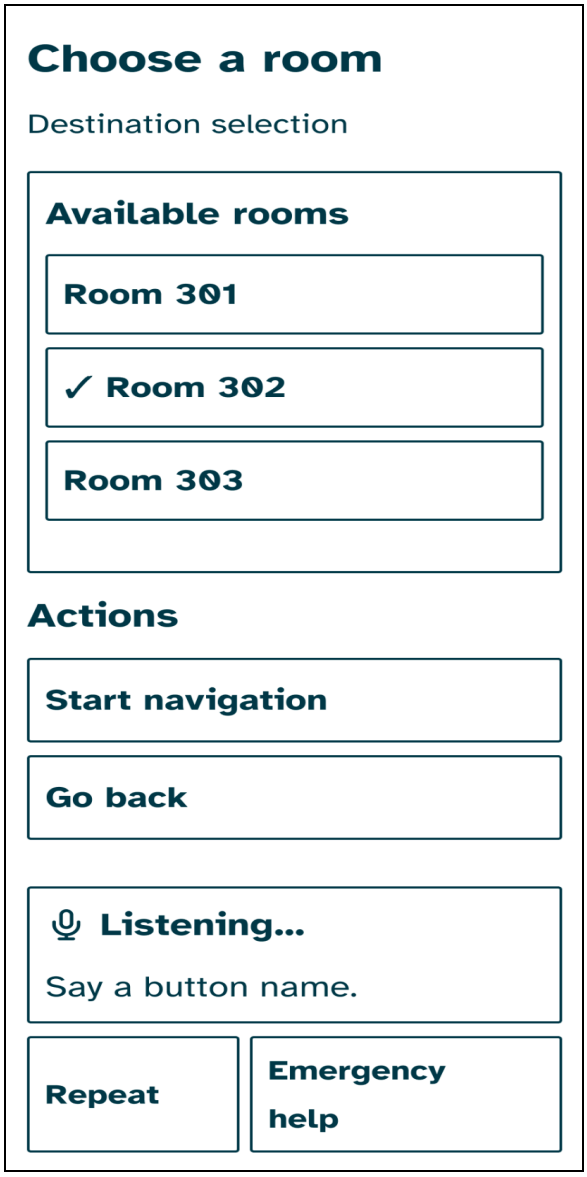
4.1.1 Home Screen



The home screen introduces Theia and asks where the user wants to go. "Speak destination" begins a voice activation destination entry while "Browse Rooms" opens the list of available rooms. The microphone icon and "Listening" label indicate that the app is accepting speech. "Repeat" repeats the current prompt and "Emergency Help" provides access to emergency

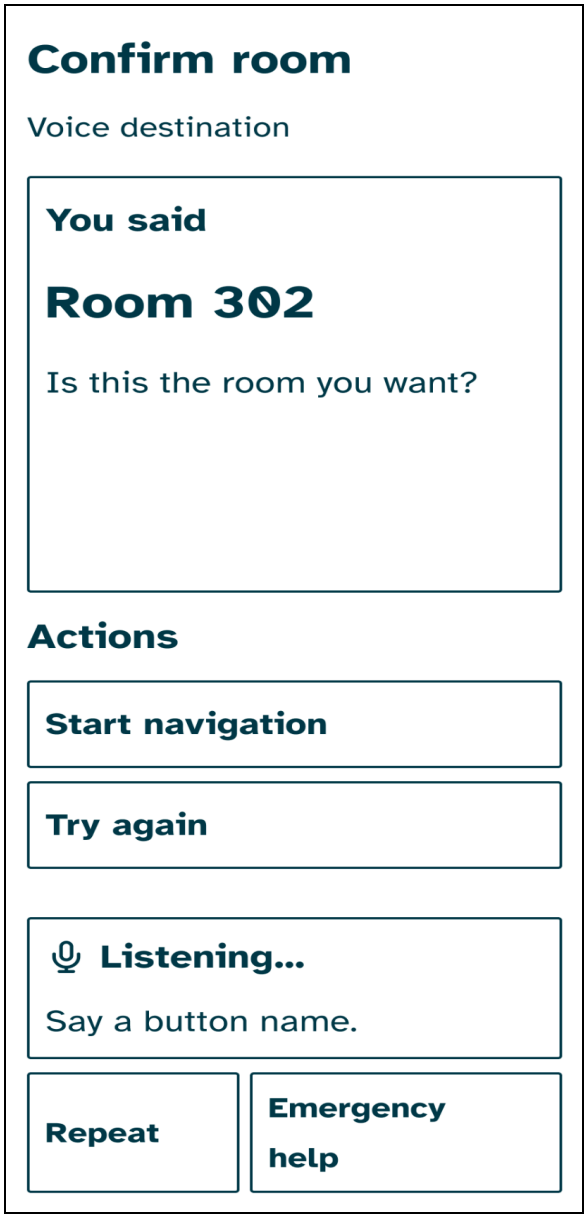
assistance.

4.1.2 Destination Selection



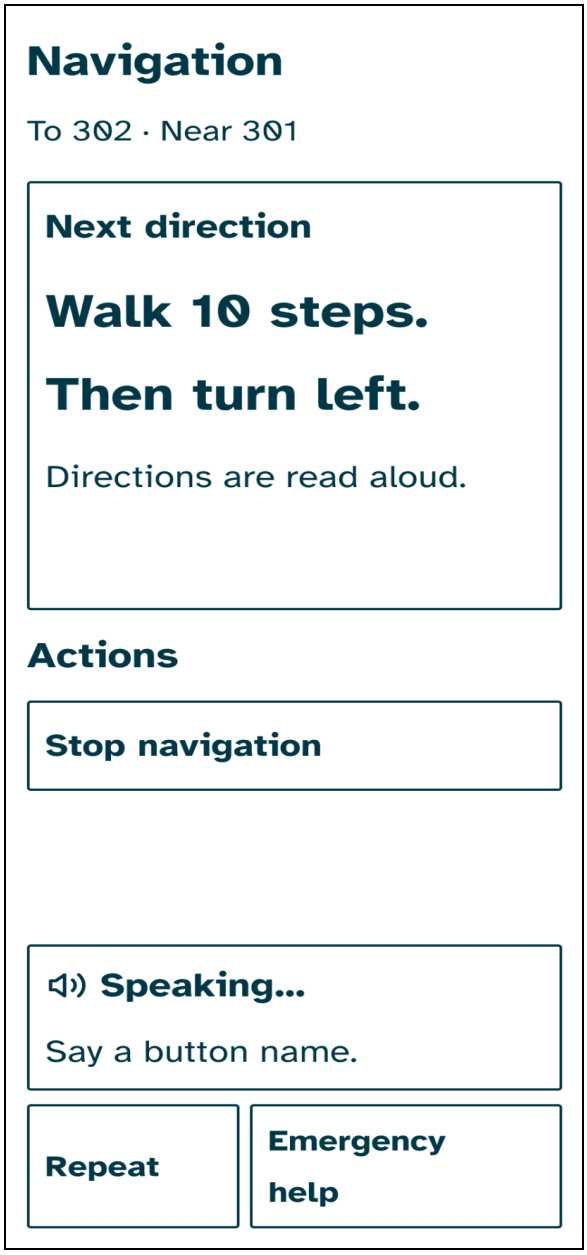
The Destination Selection screen displays the predefined rooms available for navigation. The user can select a room through touch or speech, and a checkmark identifies the selected destination. “Start navigation” proceeds to location identification, while “Go Back” returns to the Home Screen.

4.1.3 Voice Destination Confirmation



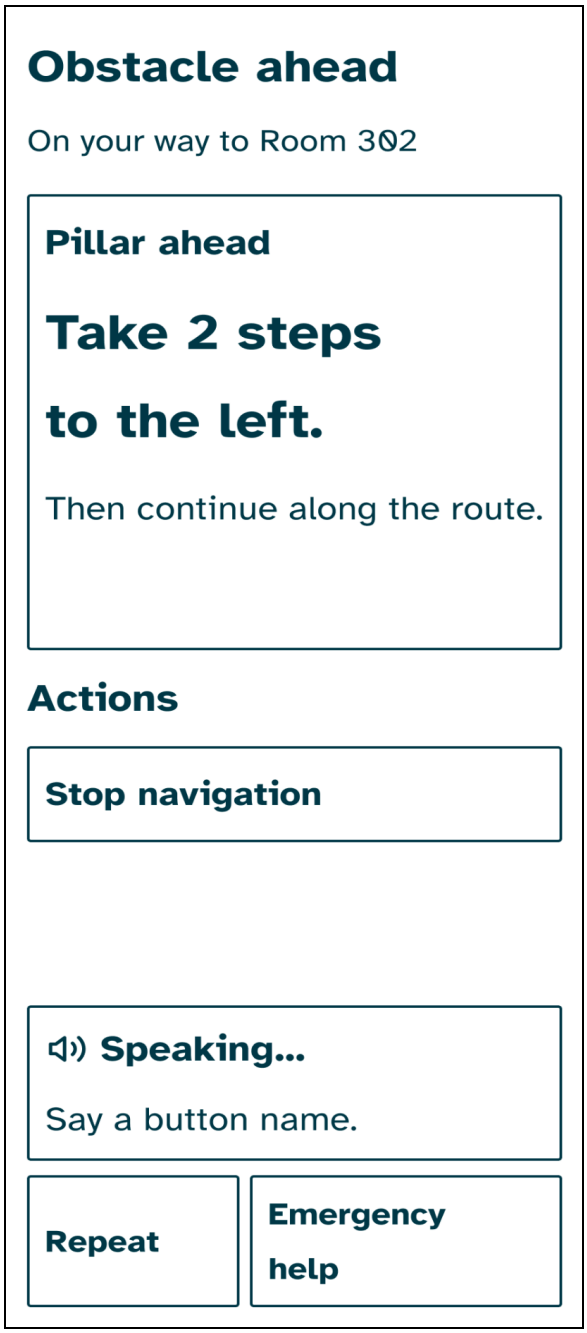
The voice Destination Confirmation screen displays the room recognized from the user's speech and asks whether it is correct. The user selects or says “Start navigation” to confirm the destination. If the recognized room is incorrect, “Try again” allows users to repeat the destination before navigation begins.

4.1.4 Active Navigation



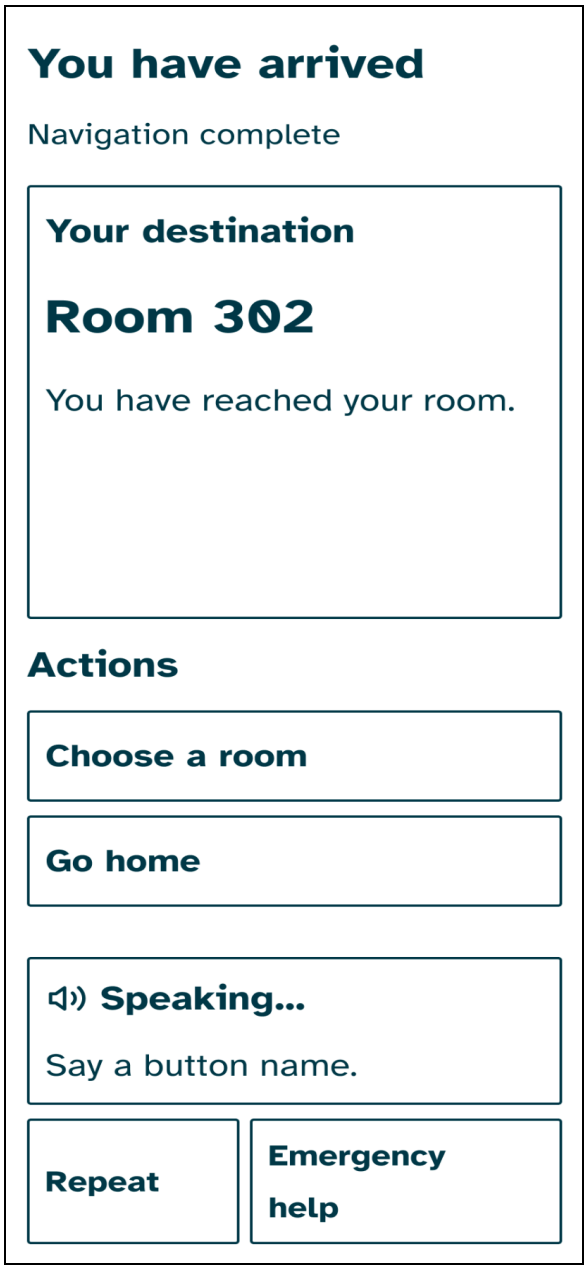
The active Navigation screen displays the destination, the user's simulated current location, and the next route instruction. Directions such as "Walk 10 steps, then turn left" are displayed and spoken aloud. Guidance updates as new simulated beacon signals indicate progress along the route. "Repeat" plays the current instruction again, while "stop navigation" end the session.

4.1.5 Obstacle Guidance



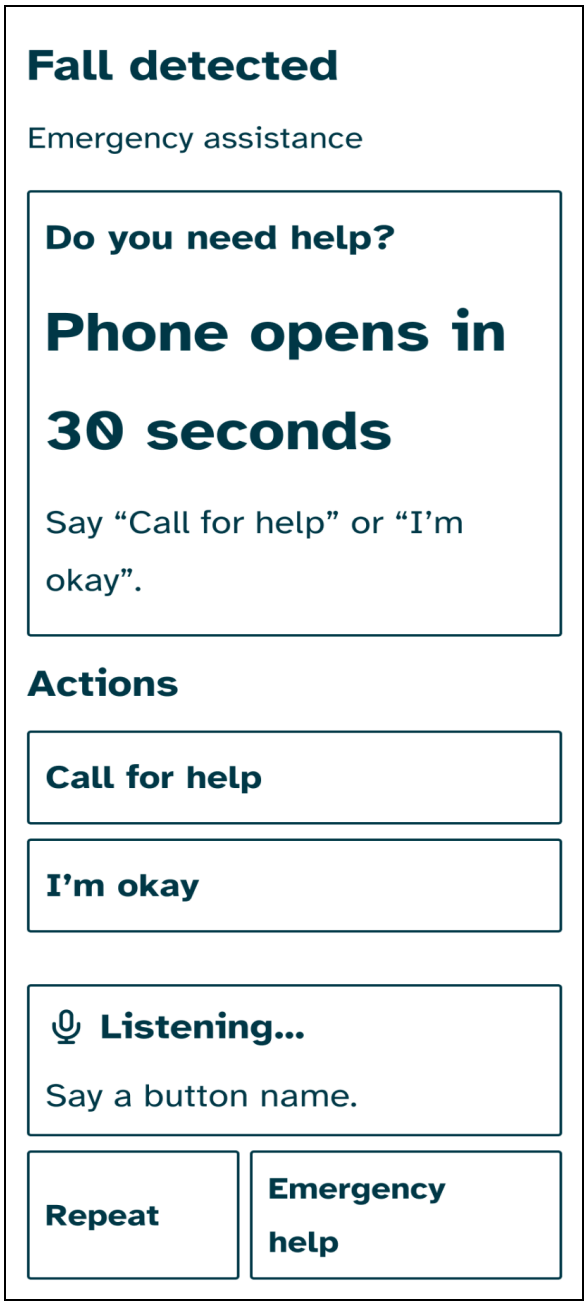
The Obstacle Guidance screen presents an instruction for avoiding a known static obstacle along the predefined route. In the illustrated example, the app identifies a pillar ahead and directs the user to take two steps to the left. The instruction is displayed and spoken aloud. This mockup represents guidance based on known obstacle information rather than verified detection of unexpected or moving objects.

4.1.6 Arrival Confirmation



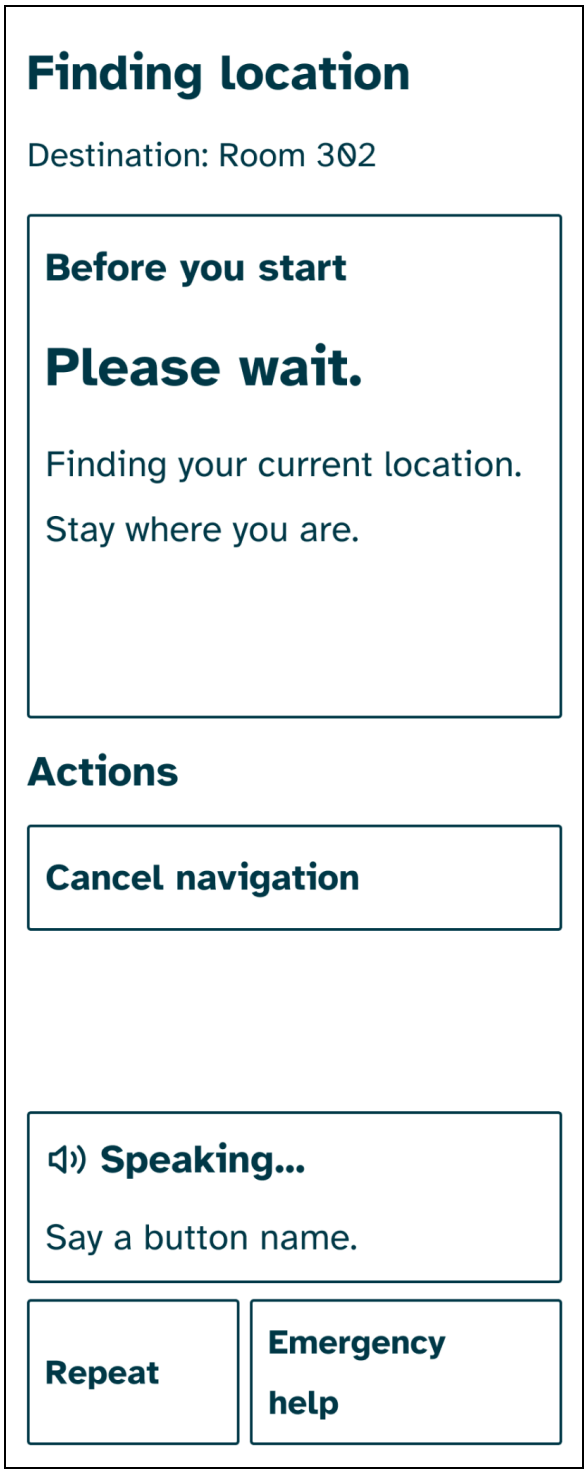
The Arrival Confirmation screen identifies the destination and announces that the user has arrived. This screen marks the completion of the current navigation session. “Choose a room” opens destination selection for another journey, while “Go Home” returns to the Home Screen.

4.1.7 Fall Detection and Emergency Assistance



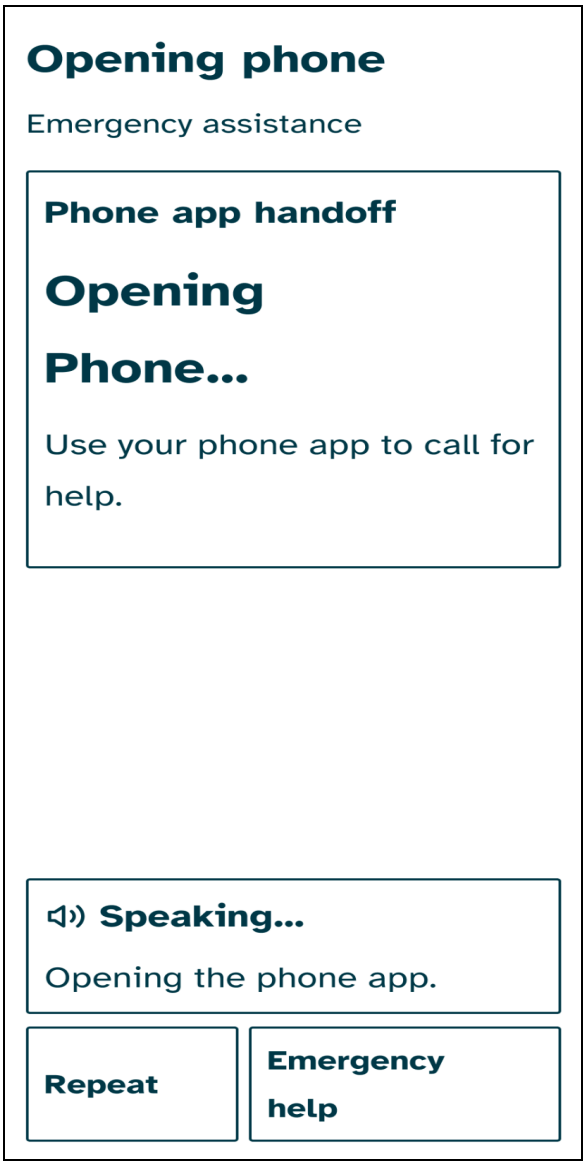
The Fall Detection and Emergency Assistance screen represents the apps response to a simulated fall. It asks whether the user needs help and displays a 30-second countdown. The user can select or say “Im okay” to dismiss the alert, or “Call for help” to begin the emergency calling handoff. If the user does not respond before the countdown ends, the proposed workflow proceeds to the same handoff.

4.1.8 Finding Current Location



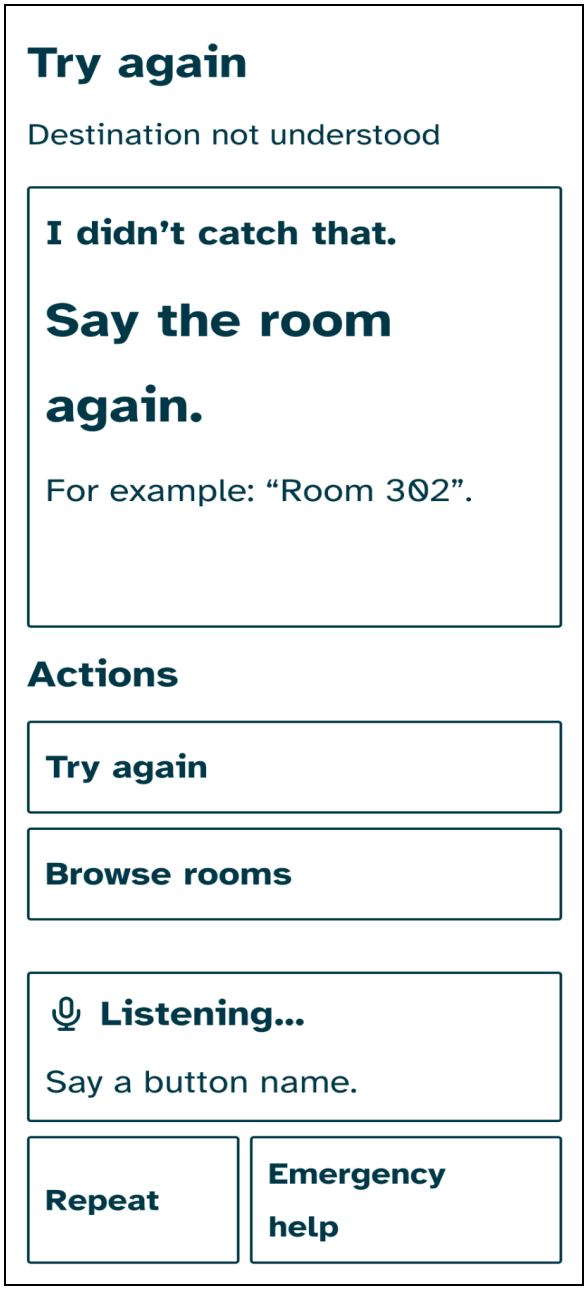
The Finding Current Location screen appears after the user confirms a destination. It asks the user to remain in place while Theia identifies the starting location using simulated beacon signals. Once a location is established, the app proceeds to Active Navigation. “Cancel navigation” cancels this process and returns to destination selection.

4.1.9 Emergency Calling Handoff



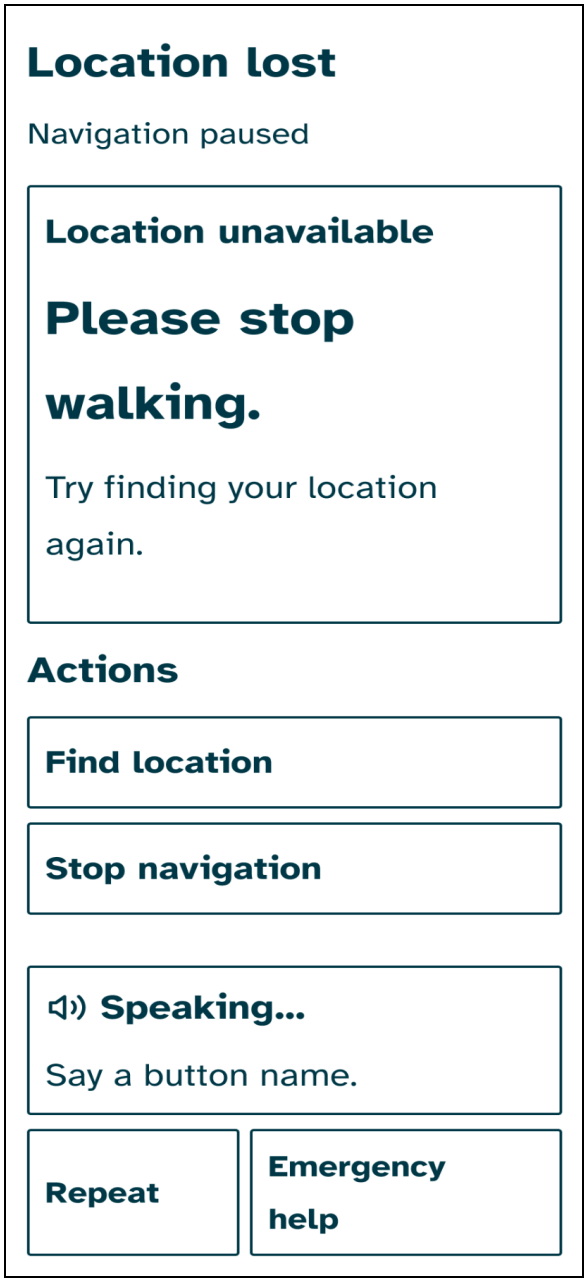
The Emergency Calling Handoff screen explains that Theia is transferring the emergency calling task to the phone's native Phone app. It follows a request for help or the expiration of the fall alert countdown. The native calling interface handles any subsequent confirmation and calling controls. This mockup illustrates the proposed transition and does not demonstrate a live emergency call.

4.1.10 Voice Input Not Understood



The Voice Input Not Understood screen informs the user that the spoken destination could not be recognized. It prompts the user to say the room again and provides an example room number. “Try again” repeats voice entry, while “Browse rooms” offers an alternative way to select a destination. The app does not begin navigation until a destination has been selected and confirmed.

4.1.11 Location Signal Unavailable.



The Location Signal Unavailable screen represents a loss of usable location information during navigation. Theia pauses guidance and instructs the users to stop walking. “Find location” retries location identification, while “Stop navigation” end the session. Route guidance resumes only after the simulated location is established again.

4.2 Speech Commands and Audio Feedback

Theia's proposed interface allows users to select destinations and activate controls through speech. Users can say the name of an available button to perform its action. A microphone icon with a “Listening” label indicates that the app is accepting speech, while a speaker icon

with a “Speaking” label indicates that the app is providing audio feedback.

A user can do any of the following commands through speech.

4.2.1 Speech Commands

Command	Intended Response
“Room 302”	Selects the room from the room list or identifies it as the proposed destination during voice input
“Browse Rooms”	Opens the destination selection screen and reads the available room choices aloud.
“Start Navigation”	Confirms the selected destination and begins identifying the user's location using simulated beacon signals.
“Try Again”	Allows the user to repeat a destination that was misunderstood or incorrectly recognized.
“Repeat”	Repeats the current instruction or screen prompt.
“Go Back”	Returns from destination selects to the Home screen
“Go Home”	Returns to the Home screen after arrival
“Choose A Room”	Opens destination selection after arrival.
“Cancel Navigation”	Cancels location identification before guidance begins and returns to destination selections
“Find Location”	Attempts to identify the user's location again after a signal loss.
“Stop Navigation”	Ends the current navigation sessions and returns to the Home screen.

“Call for Help”	Initiates the proposed handoff to the phone's native calling interface.
“Im Ok”	Dismisses the fall alert and cancels its emergency countdown.
“Cancel”	Dismisses the manual emergency assistance prompt and returns to the previous screen.

Theia provides any of the following spoken prompts to users to first explain available actions, confirm selections, and guide users along their route.

4.2.2 Audio Feedback

Situation	Example of spoken message.
Destination Selection	“Where would you like to go? Say a location or choose browse Locations.”
Destination Confirmation	“I heard Room 302. Say ‘Start navigation’ to continue or ‘Try again’ to choose another room.”
Speech not understood	“I didn't catch that, Please say the room again.”
Location identification	“Finding your current location. Please stay where you are.”
Navigation guidance	“Walk ahead 10 steps, then turn left.”
Obstacle Guidance	“Take two steps to the left to avoid the pillar."
Arrival	“You have arrived at Room 302.”
Location Signal Unavailable.	“Your location is unavailable. Please stop walking and say ‘find location’ to try again.”
Fall Detected	“A fall was detected. Say ‘Call for help’ or ‘I'm okay’. If you do not respond, the

	emergency calling handoff will begin in 30 seconds.”
Emergency calling handoff	“Open the phone's calling interface for emergency assistance.”

When speech is not understood, the app prompts the users to repeat the input. When a destination is recognized, the app asks for confirmation before beginning navigation. These interactions are represented by the preliminary mockups.

4.3 User Manual

Theia helps blind and low-vision users navigate between supported indoor locations. It provides spoken directions, guidance around known fixed obstacles, and access to emergency assistance.

You can activate available controls by touch or by saying the button name. Select Repeat to hear the current prompt or instruction again.

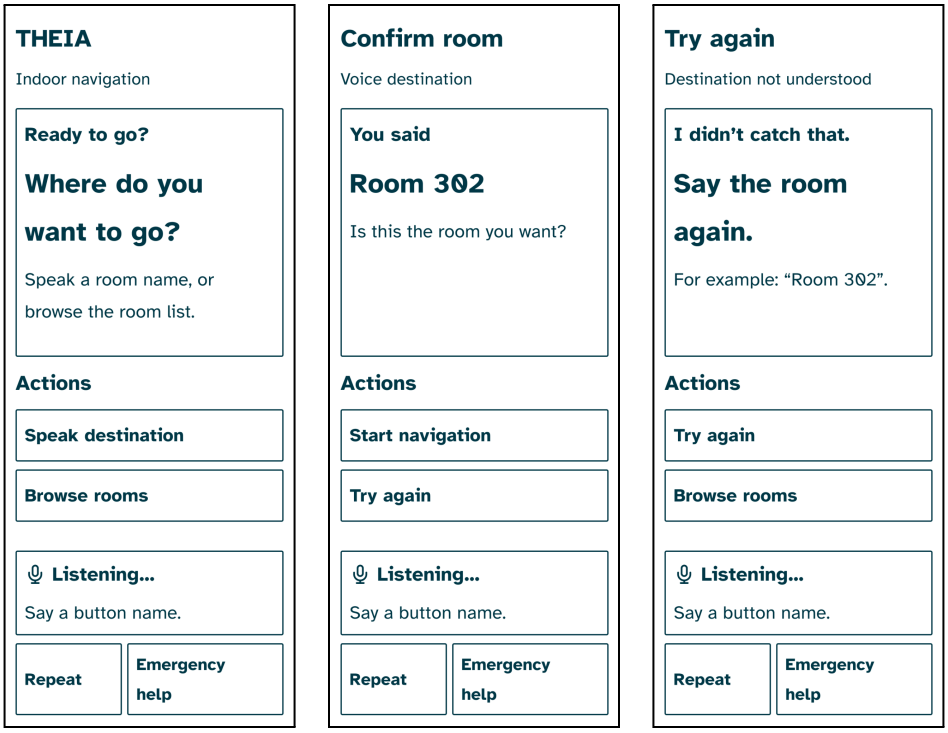
Open Theia

1. Open Theia on your Iphone.
2. Listen to the Home screen prompt.
3. Select speak destination to enter a destination by voice, or browse rooms to choose from the available destinations

Choose a destination by voice

1. Select Speak destination.
2. Say the destination, such as “Room 302.”
3. Listen as Theia repeats the recognized destination.
4. If the destination is correct, select Start navigation.
5. If it is incorrect, select Try again and repeat the destination

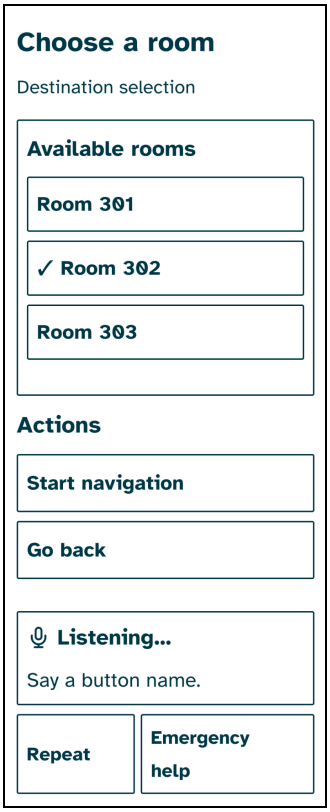
If Theia cannot understand your destination, select Try again or Browse rooms.



Voice entry: open Speak destination, confirm the recognized room, or retry an unrecognized destination.

Choose a destination from the room list

1. Select browse rooms.
2. Select your destination from the available rooms.
3. Select start navigation.



Choose a destination from the available rooms.

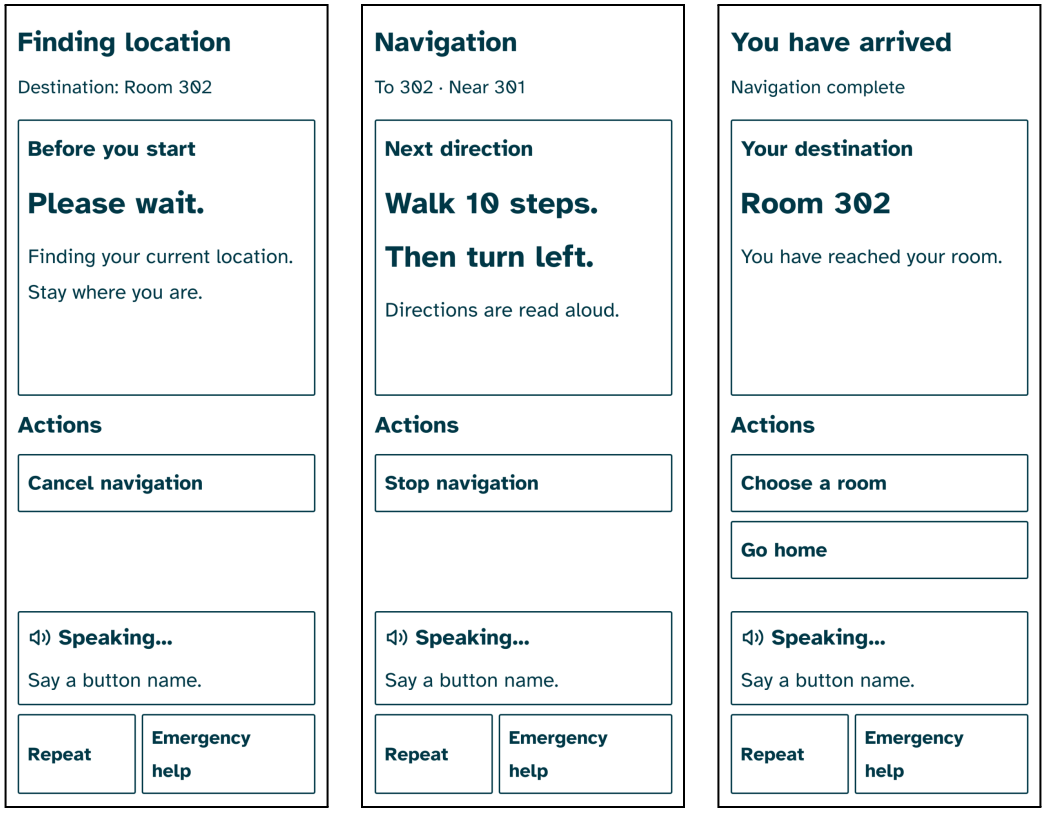
To return to the home screen before starting, select go back.

Follow directions to your destination

1. Remain in place while Theia identifies your starting location.
2. Listen to the first direction, such as “Walk 10 steps. Then turn left.”
3. Follow the instructions when it is safe to proceed.
4. Continue listening for directions as you move along the route.
5. Select Repeat wherever you need to hear the current instruction again.
6. Continue until Theia announces that you have arrived.

If Theia reports that no route is available, choose another destination.

After arrival, select Choose a room to begin another journey or Go home to return to the Home screen.



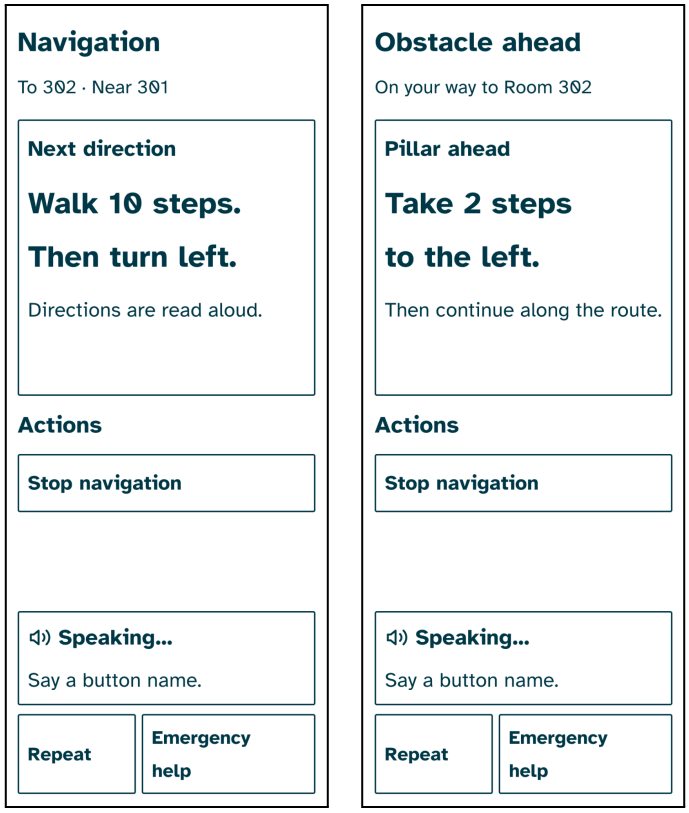
Navigation sequence: identify your starting location, follow directions, and receive arrival confirmation.

Follow guidance around a known obstacle

When your route includes a known fixed obstacle, Theia announces an instruction, such as “Take two steps to the left.”

1. Listen to the obstacle instruction
2. Select Repeat if you need to hear it again.
3. Follow the instructions when it is safe.
4. Continue listening for the next route direction.

Continue using your usual mobility aid and awareness of your surroundings while following Theia’s guidance.

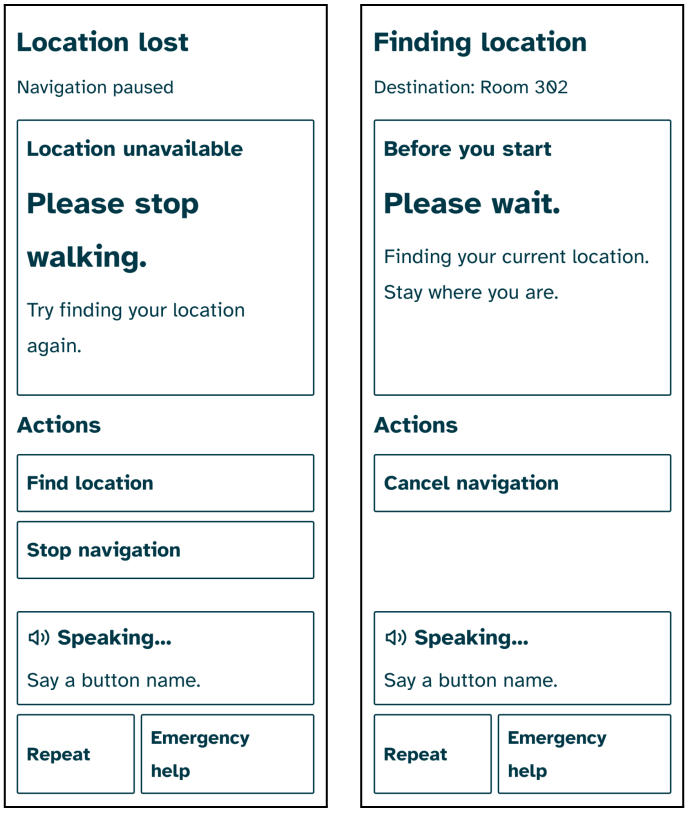


Obstacle guidance: follow the route, then follow the instruction for the known fixed obstacle.

Recover when your location is unavailable

1. Stop walking when Theia announces that your location is unavailable.
2. Select Find Location
3. Remain in place while Theia identifies your location.
4. Resume your journey when Theia provides route guidance again.

If your location remains unavailable, select Stop navigation to end the session.



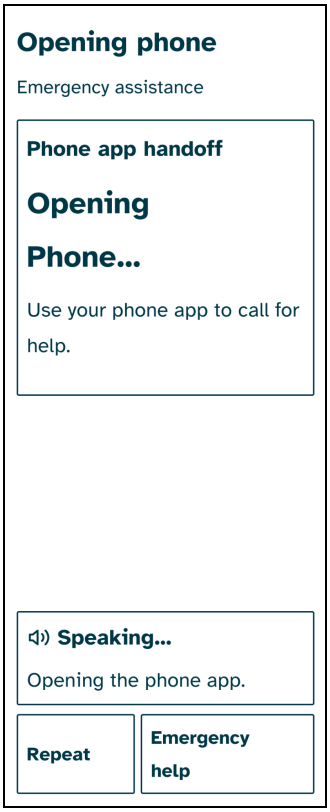
Location recovery: stop when location is unavailable, then use Find location to identify your position again.

Request Emergency assistance

1. Select Emergency help.
2. Listen to the assistance prompt
3. Select Call for help to open your iPhone's Phone app.
4. Complete the calling steps in the Phone app.

To dismiss the assistance prompt before requesting help, select Cancel.

Opening the phone app does not mean that a call has connected.

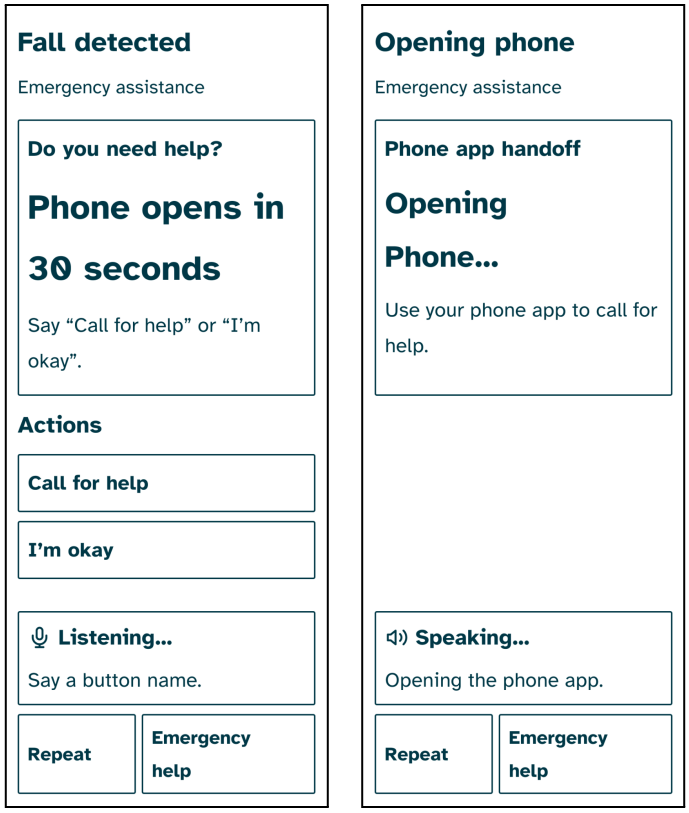


Emergency assistance: Theia opens the iPhone, where you complete the calling steps.

Respond to a fall alert

When Theia identifies a possible fall, it asks whether you need help and displays a 30-second countdown. If you do not respond before the countdown ends, Theia opens the Phone app. Complete the calling steps in the Phone.

- If you do not need assistance, select or say I'm okay to dismiss the alert and cancel the countdown.
- If you need assistance, select or say Call for help to open the Phone app, then complete the calling steps there.



Fall assistance sequence: a help request or an unanswered 30-second countdown opens the phone.

Cancel or stop navigation

- While Theia is finding your starting location, select Cancel Navigation to return to destination selection.
- During route guidance, select Stop navigation to end the session and return to the Home screen.

5. Traceability

The following mappings connect real-world problems (P), stakeholder goals (G), functional requirements (FR), and non-functional requirements (NFR).

P-01 → G-02 → FR-06, FR-07, FR-08, FR-10 → NFR-03

P-02 → G-01, G-03 → FR-01, FR-02, FR-03, FR-12 → NFR-01, NFR-02

P-03 → G-02 → FR-09 → NFR-06, NFR-08

P-04 → G-04 → FR-04, FR-05, FR-07, FR-15 → NFR-03, NFR-06

P-05 → G-05 → FR-13, FR-14 → NFR-05, NFR-08

P-06 → G-06 → NFR-04

Additional functional traceability:

FR-11 → G-03 (user control over navigation)

Every specified functional and non-functional requirement traces to at least one documented stakeholder goal or problem. In the reverse direction, every identified problem has a corresponding goal and requirement.

Prototype screens 01–11 provide preliminary demonstrations of destination selection, route guidance, obstacle warnings, arrival, location recovery, and emergency assistance. Requirements not represented by a dedicated screen, including unavailable routes and multi-floor transitions, require separate verification or further specification.

Requirements shall be reviewed against the original problems and goals whenever the prototype or project scope changes.

References

[1] Theia Project Team, “Theia – Interface Mockups,” *Figma*, 2026. [Online]. Available: https://www.figma.com/design/nUloE2z6xJFQULTEBOEz66/Theia---Interface-Mockups?node-id=0-1. Accessed: Oct. 18, 2026.

Appendix A: Phase I Plan Revisions

This appendix documents clarifications identified during requirements analysis, scenario development, and prototype review. The original project responsibilities and deadlines remain unchanged.

A.1 Scope Clarifications

The Phase I prototype will demonstrate navigation between predefined indoor destinations using simulated wireless beacon signals.

Known static obstacles will be represented through predefined building route information. Moving obstacle detection, navigation within individual rooms, and deployment of physical beacons remain outside Phase I scope.

Voice interaction, destination confirmation, navigation recovery, and emergency assistance will be represented through the preliminary interface mockups.

Multi-floor navigation remains an identified capability requiring additional specification before implementation.

A.2 Requirements and Prototype Alignment

The WRS establishes measurable requirements for simulated location updates, route availability, speech interaction, accessibility, and emergency assistance.

The prototype and user manual must describe the same behaviors and distinguish simulated functionality from actual deployed capabilities.

Emergency assistance will be demonstrated through mock interactions. A help request or an unanswered 30-second fall-alert countdown opens the native Phone app. The user completes subsequent calling steps there. Automatically placing a live emergency call, beyond opening Phone, remains outside Phase I validation.

A.3 Verification Responsibilities

Danbi Kim: Integrate updated functional and non-functional requirements and maintain traceability.

David Del Val Fraire: Review prototype interactions and accessibility requirements.

Johanna Siebol: Verify scenario and user manual consistency with the approved requirements.

Seth Trimberger: Verify document versions, consistency, and submission completeness.

Alex Shaw: Coordinate final review, resolve outstanding decisions with the team, and maintain the revised project plan.

A.4 Additional Project Risks

Risk: Navigation instructions may be incorrect when location information is unavailable.
Mitigation: Pause guidance and require location recovery before continuing.

Risk: Users may misunderstand simulated safety functions as production-ready.
Mitigation: Explicitly identify simulation limitations in the WRS, prototype descriptions, and user

manual.

Risk: Voice commands may be misunderstood.
Mitigation: Provide destination confirmation, retry options, and accessible manual controls.

Risk: Emergency calling behavior may be inconsistent across deliverables.
Mitigation: Document the selected prototype behavior and maintain unresolved production behavior as an open decision.

A.5 Change Management

The team will review changes against the WRS, prototype, scenarios, and project plan. Any material change to scope, responsibilities, or acceptance criteria must be recorded in the appropriate revision history.

The project remains scheduled for the October 18, 2026 Phase I final submission.